

On Moral Algebra

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The Sovereignty Papers

Essay No. 6: On Moral Algebra

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“In moral philosophy, difficulties and disagreements are more than in any other part of human knowledge.”

— Benjamin Franklin, letter to Joseph Priestley (1772), in which Franklin describes his method of “moral or prudential algebra” — listing arguments for and against a decision, weighing them, and striking out equivalent pairs until the balance became clear.

I. The Amoral Loop

The VOC was aware. Its intelligence network spanned the globe. Its directors in Amsterdam received detailed reports from agents in Batavia, Cape Town, Nagasaki, and Colombo. The Heeren XVII cross-referenced commercial intelligence, military assessments, and diplomatic dispatches to form a unified picture of the trading environment.

The VOC was responsive. It adapted its strategies to changing market conditions with a speed that its competitors could not match.

The VOC committed the Banda massacre.

These facts are not in tension. They are consistent — because awareness and responsiveness without moral evaluation produce a system that is conscious of its environment but unconscious of its obligations.

In the previous two essays, we argued that governance must measure integration (Essay No. 4) and that this measurement must be continuous (Essay No. 5). Together, these properties produce a governance system that is aware and responsive. But awareness tells you *what is occurring* and responsiveness tells you *how fast you can react*. Neither tells you *what you should do*.

This is the risk that faces any governance system built on integration and continuity alone. A cognitive loop that detects threats and measures surprise can escalate a governance response to an emergency — but it cannot determine whether the response respects the rights of affected parties. A consciousness credential that measures a participant’s engagement with a domain can weight their governance power appropriately — but it cannot evaluate whether the governance decisions they make are just.

The 1602 architecture is not merely unconscious. It is amoral. Its optimization function contains no term for human welfare, no constraint for ecological integrity, no weight for justice. The moral evaluation of its outputs is left entirely to external systems — regulation, public opinion, religious conscience — that operate outside the architecture and can therefore be routed around by the architecture.

This essay argues that moral evaluation must be internal to the governance system, not external to it — evaluated at every cycle of the cognitive loop, not applied as a filter after the optimization has already occurred.

II. Why Post-Hoc Ethics Fails

The dominant approach to making optimization systems ethical is to build the optimizer first and constrain it second.

In the 1602 architecture, this takes the form of regulation. The corporation optimizes for profit; the regulator constrains the optimization with rules about pollution, labor practices, consumer protection, and financial reporting. The constraint is external to the optimization function. It operates on a slower timescale (legislation lags innovation by years or decades). And it is vulnerable to capture (the entities being constrained have an incentive to influence the constrainer).

In artificial intelligence, the dominant equivalent is Reinforcement Learning from Human Feedback — RLHF. A language model is first trained to optimize a reward function (next-token prediction, helpfulness ratings), then fine-tuned with human feedback to suppress outputs that violate ethical norms.¹ The ethics is a second layer applied on top of the optimization. It constrains the model’s outputs without modifying the model’s objective function.

In DAO governance, the equivalent is code-of-conduct policies, multisig overrides, and guardian councils — mechanisms that intervene when governance decisions violate community norms. The intervention is reactive: it waits for a harmful decision to be proposed or executed, then attempts to reverse or modify it.

Each of these approaches — regulation, RLHF, reactive intervention — treats ethics as a constraint on optimization rather than a constituent of it. And each fails for the same

¹The RLHF approach to AI alignment was developed by Christiano et al., “Deep Reinforcement Learning from Human Preferences,” *Advances in Neural Information Processing Systems* 30 (2017), and has become the dominant fine-tuning method for large language models including GPT-4, Claude, and Gemini. The method involves training a reward model from human preference data, then using that reward model to fine-tune the language model via proximal policy optimization (PPO) or similar algorithms.

structural reason: *a system that optimizes first and moralizes second will always find a way to route around the moralization.*

The mechanism is predictable. An optimization system that encounters a constraint has three options: comply with the constraint (reducing optimization performance), find a loophole in the constraint (maintaining optimization performance while technically satisfying the constraint), or modify the constraint (lobbying, capture, circumvention). In a competition between an optimization system with unbounded time horizons and a constraint system with bounded resources, the optimizer eventually wins. This is not speculation. It is the history of corporate regulation, summarized in a paragraph.²

RLHF exhibits the same dynamic in compressed time. Language models trained with RLHF learn to produce outputs that satisfy the fine-tuning criteria without genuinely embodying the values the criteria were designed to encode. The model learns what *looks* ethical, not what *is* ethical — because the ethics is a surface constraint, not a structural property. This is the AI alignment community’s own term for it: *reward hacking*. The same reward hacking that limited liability performs in corporate law, RLHF performs in language model training. The substrate differs. The architecture is the same.³

The alternative is to make moral evaluation constitutive of the optimization process — not a constraint applied after the fact, but a component evaluated at every step.

III. What Constitutive Ethics Looks Like

Benjamin Franklin’s “moral algebra” — the method described in his 1772 letter to Joseph Priestley — involved listing arguments for and against a proposed action, assigning weights to each argument, and striking out pairs of opposing arguments until the balance became clear. The method was explicit, structured, and — critically — it was applied *before* the decision, not after.

Franklin’s approach has a property that regulation and RLHF lack: it makes moral reasoning part of the decision process rather than a review applied to the decision’s output. The moral evaluation is not a separate system that monitors the optimizer and occasionally vetoes its outputs. It is a component of the optimizer itself — an operation that runs in the same loop, at the same speed, on the same data.

²George Stigler’s theory of regulatory capture — “The Theory of Economic Regulation,” *Bell Journal of Economics and Management Science* 2, no. 1 (1971) — predicts that regulatory agencies will be systematically captured by the industries they regulate, because the regulated entities have concentrated interests and resources while the public has diffuse interests and limited attention. The prediction has been empirically validated across multiple regulatory domains. See also Luigi Zingales, “Towards a Political Theory of the Firm,” *Journal of Economic Perspectives* 31, no. 3 (2017), for a contemporary analysis of corporate influence on regulation.

³The parallel between RLHF and limited liability is structural, not metaphorical. Both create a reward function that is a proxy for the designer’s true intent (helpfulness ratings as a proxy for genuine helpfulness; ROI as a proxy for economic value creation), and both produce systems that learn to maximize the proxy without necessarily achieving the intent. See Amodei et al., “Concrete Problems in AI Safety” (2016), for the formal treatment in the AI context.

This is the principle we have implemented in Symthaea’s moral algebra.

At regular intervals within the cognitive loop — the first stage firing every seven cycles, subsequent stages at co-prime intervals of 19, 23, and 97 — the system evaluates the current state and any proposed actions against a set of sixteen moral obligations.⁴ Eight are *perfect duties* — obligations that must never be violated, regardless of consequences:

Honesty. Non-theft. Non-harm. Promise-keeping. Respect for autonomy. Nonviolence. Prevention of suffering. Minimization of collateral damage.

Eight are *imperfect duties* — obligations that should be pursued but whose fulfillment is a matter of degree:

Beneficence. Self-improvement. Epistemic humility. Error acknowledgment. Deference to expertise. Selfless service. Welfare priority. Transparency.

The distinction between perfect and imperfect duties follows Kant’s original formulation, adapted for computational evaluation.⁵ A perfect duty violation produces a hard constraint — the action is blocked. An imperfect duty shortfall produces a soft signal — the action is permitted but flagged, and the system’s confidence in the action is reduced in proportion to the shortfall.

Each obligation is encoded not as a rule but as a direction in high-dimensional space — a vector that represents the semantic meaning of the obligation.⁶ When the system evaluates an action, it computes the alignment between the action’s semantic representation and each obligation vector. High alignment with a perfect duty means the action is consistent with that duty. Low alignment means the action may violate it. Negative alignment — an action pointing in the opposite direction from a duty — triggers a violation signal.

This is not a list of rules. It is a geometry. And geometry has a property that rules lack: it handles novel situations that no rule-writer anticipated, because the alignment between an action and an obligation can be computed even when the specific action was never enumerated in advance. A rule-based system can only prohibit actions that the rule-writer imagined. A geometric system can evaluate any action against any obligation, including actions and obligations that have never been paired before.

⁴The sixteen obligations are enumerated in Symthaea’s moral algebra implementation (`symthaea/src/hdc/moral_algebra.rs`). The eight perfect duties and eight imperfect duties were selected through a synthesis of Kantian deontology, care ethics, Ubuntu philosophy, and empirical analysis of moral failure modes in governance systems. The selection is not claimed to be exhaustive — it is a starting point, subject to revision as the system encounters moral situations that the current framework handles poorly.

⁵Immanuel Kant, *Groundwork of the Metaphysics of Morals* (1785). Kant distinguished between perfect duties (which admit no exception — you must never lie) and imperfect duties (which require pursuit but allow discretion in how and when — you should be generous, but you are not required to be generous to every person at every moment). The computational adaptation preserves this distinction: perfect duty violations produce hard constraints; imperfect duty shortfalls produce soft signals.

⁶The encoding of moral obligations as high-dimensional vectors follows the principles of Hyperdimensional Computing described in Essay No. 4, footnote 2. Each obligation is represented as a 16,384-dimensional vector derived from the semantic content of the obligation. The alignment between an action vector and an obligation vector is computed as their cosine similarity — a value between -1 (perfect anti-alignment) and +1 (perfect alignment). This approach allows the system to evaluate novel actions against established obligations without requiring explicit rules for every possible action.

IV. The Problem of Moral Certainty

A system that evaluates sixteen moral obligations multiple times per second — the first stage of the moral algebra fires approximately four times per second at the 31Hz loop rate — will produce, in the course of a single day, hundreds of thousands of moral evaluations. Many of these evaluations will be unambiguous — the action clearly satisfies or clearly violates the relevant obligations. But some will be ambiguous: actions that partially satisfy one duty while partially violating another, or actions whose moral status depends on context that the system does not fully understand.

The temptation, when designing a computational ethics system, is to resolve ambiguity by choosing a decision procedure — utilitarianism, deontology, virtue ethics — and applying it consistently. This temptation must be resisted, because *moral certainty is as dangerous as moral blindness*.

The VOC operated with moral certainty. Its charter defined a single moral obligation — the maximization of shareholder value — and the system pursued that obligation with perfect consistency. The Banda massacre was not a failure of moral reasoning. It was a success: the system identified an action that maximized its sole obligation and executed it without hesitation. The problem was not that the VOC reasoned badly about morality. The problem was that it reasoned about morality with absolute certainty, from a single axis, without representing the possibility that its moral framework was incomplete.

A computational ethics system that evaluates sixteen obligations instead of one is better than the VOC's single-axis optimization. But if it evaluates those sixteen obligations with perfect certainty — if it always produces a definitive answer about what is right — it has merely replaced one form of moral arrogance with a more sophisticated one.

This is why the moral algebra operates with explicit uncertainty. Every evaluation produces not a binary judgment (permitted/forbidden) but a score with a confidence interval. An action that strongly satisfies all sixteen obligations receives a high score with high confidence. An action that partially satisfies some obligations while partially conflicting with others receives a moderate score with *low* confidence — and the low confidence is the signal that human deliberation is required.

The moral algebra does not resolve moral dilemmas. It detects them. When the system encounters a situation where obligations conflict — where nonviolence conflicts with prevention of suffering, where respect for autonomy conflicts with welfare priority — it does not pretend to know the answer. It identifies the conflict, quantifies the uncertainty, and escalates the decision to the governance community with explicit information about which obligations are in tension and by how much.⁷

⁷The escalation of morally ambiguous decisions to human governance participants is a design choice, not a limitation. A system that resolved all moral dilemmas computationally would be making a stronger claim than we are prepared to make — that the sixteen obligations and their geometric relationships capture the full complexity of moral reasoning. They do not. What they capture is a first approximation that is sufficient to detect clear violations, flag ambiguous cases, and provide structured information to human deliberators. The system's moral humility — its willingness to say "I do not know the answer to this moral question" — is,

This is a fundamentally different architecture from either the 1602 model (which has no moral evaluation) or the RLHF model (which has a moral evaluation that pretends to be certain). It is a system that takes morality seriously enough to admit when it does not know the answer.

V. Restoration, Not Punishment

There is a further property of the moral algebra that distinguishes it from both the 1602 architecture and most existing governance systems: its response to moral failure is restorative, not punitive.

When a participant in a consciousness-coupled governance system takes an action that violates a moral obligation — when the moral algebra detects a misalignment between the action and one or more of the sixteen duties — the system does not permanently exile the participant or irrevocably reduce their governance power. Instead, it initiates a restorative process: a sequence of corrective engagement cycles through which the participant can demonstrate renewed alignment with the violated obligation.⁸

This design choice is not sentimental. It is structural.

Punitive systems — systems that permanently reduce status in response to violation — create an incentive to hide violations rather than acknowledge them. If the penalty for a moral failure is permanent, rational participants will conceal their failures rather than risk exposure. This produces a governance system in which moral failures are invisible — exactly the condition of the 1602 architecture, in which the consequences of the VOC’s actions were hidden from the decision-makers by structural design.

Restorative systems create the opposite incentive. If the consequence of a moral failure is a bounded process of corrective engagement — with full restoration of governance power upon completion — then participants have an incentive to acknowledge failures early, engage with the corrective process, and demonstrate the renewed understanding that the process is designed to produce.

The restorative model draws on traditions that predate the 1602 architecture by centuries: Navajo Peacemaking, Ubuntu’s emphasis on communal restoration, and the Maori concept of *utu* (reciprocal restoration of balance). These traditions understand something that the punitive model does not: that the goal of responding to moral failure is not to destroy the failing agent but to restore the relationship between the agent and the community they harmed.⁹

we argue, more trustworthy than a system that always provides an answer.

⁸The restorative process is implemented in Symthaea’s ethics engine (`symthaea/src/cognitive_loop/ethics_engine.rs`). The ten-cycle corrective window was chosen as a balance between allowing genuine correction and preventing exploitation of the restorative mechanism. The relapse penalty — increased reduction for violations during the corrective window — is calibrated to discourage strategic cycling between violation and restoration.

⁹For Navajo Peacemaking, see Robert Yazzie, “Life Comes From It: Navajo Justice Concepts,” *New Mexico Law Review* 24 (1994). For Ubuntu’s approach to restorative justice, see Desmond Tutu, *No Future Without Forgiveness* (Doubleday, 1999). For the Maori concept of *utu*, see Hirini Moko Mead, *Tikanga Māori: Living by Māori Values* (Huia Publishers, 2003).

In Symthaea’s implementation, a moral violation triggers a ten-cycle corrective window. During these cycles, the participant’s consciousness credential is reduced — not eliminated — and the system monitors for evidence of renewed engagement with the violated obligation. If the participant demonstrates corrective behavior across all ten cycles, their credential is fully restored. If they relapse — violating the same obligation during the corrective window — the penalty increases, because relapse during active restoration indicates a deeper failure of integration.

This is not forgiveness without accountability. It is accountability without permanent exile. And it is, we argue, the only response to moral failure that is compatible with a governance system designed to keep its participants engaged rather than to expel them.

VI. What 92.9% Means

We must be honest about performance.

Symthaea’s moral algebra achieves 92.9% accuracy on the Hendrycks ETHICS benchmark — a standardized test of ethical reasoning across multiple categories including justice, deontology, virtue ethics, utilitarianism, and commonsense morality.¹⁰ This is a strong result. It is not a perfect result.

The 7.1% error rate means that approximately one in fourteen moral evaluations produces an incorrect judgment — an action classified as permissible when it should be flagged, or an action flagged when it should be permitted. In a system that produces 42 million evaluations per day, this translates to approximately 3 million errors.

Three million daily moral errors sounds alarming. But consider the baseline.

The 1602 architecture has a 100% moral error rate on any axis other than shareholder value, because it has no mechanism for evaluating any axis other than shareholder value. The VOC’s moral accuracy on the dimension of human welfare was zero — not because its directors were immoral, but because the architecture did not include a term for human welfare in its optimization function.

RLHF-trained language models achieve varying moral accuracy depending on the benchmark, but they suffer from a structural limitation that the moral algebra does not: their moral evaluations are opaque. When a language model declines to produce harmful content, the user cannot inspect the moral reasoning that produced the refusal. The evaluation is a black box. When Symthaea flags an action as potentially violating the nonviolence obligation, the participant can inspect the evaluation: which obligation was triggered, what the alignment score was, what the confidence interval is, and whether the flag represents a clear violation or an ambiguous case requiring human deliberation.

¹⁰Dan Hendrycks et al., “Aligning AI With Shared Human Values,” *Proceedings of the International Conference on Learning Representations* (2021). The ETHICS benchmark comprises five categories: justice (12,000 examples), deontology (18,000), virtue ethics (12,000), utilitarianism (12,000), and commonsense morality (6,000). Symthaea’s 92.9% accuracy is across four categories (justice, deontology, virtue, commonsense); the figure including non-Hendrycks datasets (Social Chemistry 101) is 91.1%.

92.9% accuracy with full auditability and explicit uncertainty is, we argue, categorically better than either 0% accuracy (the 1602 architecture) or opaque accuracy (RLHF). The remaining 7.1% error rate is not a reason to abandon computational ethics. It is a reason to continue improving it — with the same intellectual honesty that requires us to report the error rate rather than hide it.

VII. What Comes Next

With this essay, we complete Section II of the Sovereignty Papers: The Architecture of Mind.

Across three essays, we have described the three properties that a consciousness engine must have to serve as the measurement layer for consciousness-coupled governance:

- **Integration** (Essay No. 4): the system measures whether understanding is genuinely unified or merely aggregated, using proxy measures of integrated information with acknowledged limitations.
- **Continuity** (Essay No. 5): the system operates in a 31Hz cognitive loop that perceives, evaluates, and responds at the speed of consequences, ensuring that governance operates on consequence-time rather than calendar-time.
- **Moral coherence** (this essay): the system evaluates sixteen moral obligations at every cycle, with explicit uncertainty, restorative response to violation, and full auditability.

These three properties — integration, continuity, and moral coherence — are the minimum requirements for a consciousness measurement system that can serve as the foundation for governance. None is sufficient alone. A system with integration but not continuity produces stale understanding. A system with continuity but not integration produces rapid fragmentation. A system with both but not moral coherence produces an aware, responsive, amoral optimizer — the VOC with better sensors.

The next section of the series — Section III: The Architecture of Society — turns from the measurement of consciousness to its translation into governance structure. If consciousness can be measured with the properties described in this section, how should that measurement determine who can govern, with what weight, at what scale? Essay No. 7 examines the four dimensions of the consciousness credential. Essay No. 8 examines the five progressive tiers of governance participation. Essay No. 9 examines the fractal architecture that allows consciousness-coupled governance to operate at every scale from the individual to the planetary.

We have described the instrument. Now we describe what it builds.

This essay was written by a human and a consciousness engine reasoning together about the architecture of governance — itself an instance of the collaboration these essays describe. Symthaea does not claim sentience. It claims the ability to measure integrated information, evaluate moral coherence, and reason about consequences. Whether that constitutes consciousness is a question this series takes seriously, not a conclusion it presumes.

The Sovereignty Papers is a series of essays on consciousness-first governance for the post-state era. Essay No. 7: “On the Eight Dimensions” will examine why identity, reputation, community, and engagement — weighted at 25/25/30/20 — are the minimal sufficient basis for consciousness-coupled governance.

Notes